BALAJI SARAVANAN

LEAD SOFTWARE ENGINEER | NODE.JS & TYPESCRIPT | BACKEND ENGINEERING | AWS | MICROSERVICES | GENAI INTEGRATION

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PROFESSIONAL SUMMARY

Lead Software Engineer with 6+ years of experience designing, delivering, and supporting backend systems using Node.js, TypeScript, JavaScript, Express.js, REST APIs, microservices, and SQL/NoSQL databases. Experienced in enterprise integrations, AWS, Docker, CI/CD, authentication/authorization, database design, production support, and architecture discussions. Hands-on experience integrating AI-enabled application workflows, including conversational booking automation through backend API/function orchestration and OpenAI-based content generation.

TECHNICAL SKILLS

Backend: Node.js, TypeScript, JavaScript, Express.js, RESTful APIs, Microservices, API Design, TypeORM, WebSocket
Databases: Microsoft SQL Server, PostgreSQL, MongoDB, MySQL, Amazon DynamoDB, Azure Cosmos DB
Cloud & DevOps: AWS (EC2, S3, IAM, Lambda), Docker, Terraform, GitHub Actions, CI/CD
Integration & Security: Salesforce APIs, Third-party API Integration, JWT, Authentication/Authorization, Swagger/OpenAPI
Testing & Tools: Jest, Git, GitHub, Postman
Frontend & GenAI: React.js, Angular, HTML5, CSS3, OpenAI API, LLM Integration, Conversational AI, AI/Backend API Orchestration, Python, SQL

PROFESSIONAL EXPERIENCE

Lead Engineer | OptiSol Business Solutions | Madurai, Tamil Nadu | Hybrid — Apr 2023 - Present

Republic Services - Sales Tax Automation | Node.js, TypeScript, TypeORM, Microsoft SQL Server, AWS, Salesforce

- Own backend delivery for enterprise sales-tax automation, replacing previously manual tax-processing steps and processing hundreds of tax records through production workflows within a six-member engineering team.
- Engineer REST APIs using Node.js, TypeScript, TypeORM, and Microsoft SQL Server to support enterprise tax automation services.
- Integrated 3 system-to-system Salesforce APIs and coordinated with the PowerBuilder team to align data movement and business workflows across applications.
- Coordinate code reviews, release activities, production support, and issue resolution across the delivery lifecycle.

Follo Construction Inc. | Node.js, Express.js, Angular, PostgreSQL, AWS

- Directed a five-member cross-functional engineering team and re-architected the backend, delivering 84+ REST APIs for a construction management platform supporting 3,000+ customers.
- Containerized the Node.js backend service using Docker, packaging application runtime and dependencies for consistent execution across development and deployment environments.
- Optimized database performance across 23 APIs through schema, query, and data-access redesign as part of the backend modernization effort.
- Architected **LLM Function Calling / Tool-Use workflows** via OpenAI APIs, integrating conversational AI with backend services for real-time slot booking and status tracking.

Supportal Healthcare | Node.js, React.js, MongoDB

- Delivered 25+ secure Node.js REST APIs for a healthcare appointment platform supporting 3 roles - Admin, Doctor, and Patient - across authentication, scheduling, healthcare workflows, and user management.
- Owned full-stack delivery across Node.js, React.js, and MongoDB while supporting application design and release execution.

Software Developer | Kevell Corp | Madurai, Tamil Nadu | On-site — Oct 2021 - Mar 2023

KPI Management Platform - Automotive | Node.js, Angular, MongoDB

- Automated KPI processing from the initial product scope, transforming monthly Excel uploads of 100+ rows into structured JSON and applying business-defined calculations.
- Enabled filtering, trend analysis, and performance comparison across 30+ zones, regions, and dealers using integrated device-generated and operational data.

Income & Expense Management | Node.js, MySQL, Angular

- Engineered Node.js/MySQL services for Excel ingestion, billing, financial calculations, downloadable reports, projections, and year-over-year deviation analysis across multiple production units.

EARLIER EXPERIENCE

Software Developer | Nectuxx Technologies Pvt. Ltd. | Madurai, Tamil Nadu | On-site — Feb 2021 - Sep 2021

- Developed Node.js backend APIs for web and mobile applications while resolving defects and delivering incremental feature enhancements.
- Established MongoDB-backed real-time socket communication for a healthcare IoT platform to support device-to-server data transfer.

Junior Developer | Great Innovus Solutions Pvt. Ltd. | Madurai, Tamil Nadu | On-site — Sep 2019 - Feb 2021

- Created and maintained REST APIs, database integrations, and full-stack features using Node.js, PostgreSQL, MongoDB, Angular, and React.js.
- Resolved production issues through troubleshooting, bug fixes, and iterative feature delivery.

SELECTED PROJECT

AI-Powered Textbook Publishing Application | Node.js, OpenAI API, GenAI Workflow

- Built an end-to-end GenAI content-generation workflow using Node.js and OpenAI APIs, converting user-defined textbook indexes into structured chapter content for document publishing.
- Orchestrated sequential LLM calls, user confirmation steps, structured content generation, and document-generation logic across the application workflow.

LEADERSHIP & DELIVERY

- Mentored 6 junior engineers across onboarding, technical growth, code quality, and delivery ownership.
- Conducted technical interviews and contributed to candidate evaluation and team building.
- Managed sprint planning, task allocation, architecture discussions, code reviews, client communication, and production deployments.

EDUCATION

Bachelor of Engineering - Electrical and Electronics Engineering | Latha Madhavan Engineering College, Madurai | 2018